

PAPER_1364 — Quantum Biology Coherence in Photosynthesis

Framework: UQFF — Star-Magic v5.27+ | **Tier:** G Bio/Complex (CLOSED) **Calculator:** `calculate_paradox({"paradox": "quantum_biology"})`

Master Expression

$$T_{\text{coherence}} = \hbar \cdot \omega_{\text{SCm}} / k_B / \beta_i = 99.5 \text{ K}$$

Match: 29% from FMO ~77 K

Interpretation

SCm phonon-assisted exciton transport. FMO complex coherence at SCm phonon $\times \beta_i$ temperature scale.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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